



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MAT3006
Course title (English)	Real Analysis
Course title (Chinese)	实分析
Units	3
Language of Instruction	English
Description (English)	This course introduces Lebesgue theory of functions of one variable, covering Lebesgue measure, Lebesgue integrals and differentiation, Lpspaces, weak convergence, weak derivatives, and Sobolev inequality and Sobolev spaces.
Description (Chinese)	此课介绍实轴上的勒贝格理论, 包括勒贝格测度, 勒贝格积分以及微分理论, 弱收敛。

2. Prerequisites / Co-requisites

A. Prerequisites

MAT2060 (Mathematical Analysis Extended)

B. Co-requisites

3. Learning Outcomes

The students are expected to master basic concepts and important results in Lebesgue theory of functions, including Lebesgue measurability, Lebesgue integrals and differentiation theory. Upon completing this course, the students will be able to take courses such as functional analysis, abstract measure theory, harmonic analysis/Fourier analysis, modern partial differential equations.



4. Course syllabus

□ Text book: Real Analysis, by Halsey L. Royden and Patrick M. Fitzpatrick, 2010, 机械工业出版社 01-2010-3826

□ Topics to be covered:

- Lebesgue outer measure, Lebesgue measure, nonmeasurable sets, Cantor set
- Measurable functions, Egoroff's theorem, Lusin's theorem
- Lebesgue integrals
- Fatou's lemma, Various convergence theorems
- Differentiability, bounded variations, and absolute continuity
- L_p space, Holder's inequality, weak convergence
- Sobolev space, Sobolev inequality

5. Assessment Scheme

Component/ method	% weight
Homework	25
Mid-term test	25
Final exam	50

6. Grade descriptor

General

Grade	Description
A	Outstanding performance on all learning outcomes.
A-	Generally outstanding performance on all (or almost all) learning outcomes.
B	Substantial performance on all learning outcomes, or high performance on some learning outcomes which compensates for less satisfactory performance on others, resulting in overall substantial performance.
C	Satisfactory performance on the majority of learning outcomes, possibly with a few weaknesses.
D	Barely satisfactory performance on a number of learning outcomes.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.



7. Feedback for evaluation

Collect student feedback from CTE by the end of the course

Collect opinions from peers by internal and external course reviews

8. Reading

Required Reading

1. Real Analysis, 01-2010-3826, /, United States

9. Course components

Activity	Hours/week
Lectures	3
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
00000 00001	Lebesgue outer measure
00000 00002	Lebesgue measure
00000 00003	Lebesgue measurable sets, Lebesgue measurable functions
00000 00004	Convergence a.e., convergence in measure
00000 00005	Egorov's theorem, Lusin's theorem
00000 00006	Lebesgue integral, monotone convergence theorem
00000 00007	Fatou's lemma, Lebesgue integrals of general measurable functions
00000 00008	Lebesgue's dominated convergence theorem, various convergence results
00000 00009	Dini derivatives, differentiability of monotone functions
00000 00010	Functions of bounded variations



00000 00011	Fundamental theorem of calculus, absolutely continuous functions
00000 00012	L_p spaces, Holder's inequality, Minkowski inequality
00000 00013	Weak convergence, weak derivatives
00000 00014	Sobolev spaces, Sobolev inequality

11. Any other information

Prohibit the use of AI tools: AI tools (particularly GPT and generative AI tools) are not allowed in submitted reports. Any student who violates this rule will be considered to be engaging in academic dishonesty. The instructor may specify which AI tools are prohibited in this course.

12. Version date

Version number	4
As of (date)	2025/08/08